

Prithul Das

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Education

Ahsanullah University of Science and Technology (AUST)

B.Sc., Mechanical Engineering (ME)

April 2022 – May 2027 (Expected)

Dhaka, Bangladesh

CGPA: 3.28/4.0 (Current, up to 3rd Year 2nd Semester; Projected: $\sim 3.42/4.0$)

Relevant Coursework: Fluid Mechanics, Heat and Mass Transfer, Thermodynamics, Machine Design, Solid Mechanics, Multivariable Calculus, Differential Equations, Linear Algebra, Basic Programming, Engineering Mathematics

- **Thesis:** Final-year thesis commencing July 2026; expected focus in computational fluid dynamics or fluid–structure interaction. **Supervisor:** TBD

Research Interests

Computational Fluid Dynamics (CFD) · Aerodynamics & Aerothermodynamics · Fluid–Structure Interaction (FSI) · Transonic & Hypersonic Flow Physics · Thermal Systems · Physics-Informed Machine Learning

Research Experience

Aerothermodynamic Analysis of a Modified Atmospheric Re-entry Vehicle Heat Shield

2025 – Present

AUST Applied Modelling and Simulation Lab (AASML), Dhaka, Bangladesh

- Conducting hypersonic flow analysis at Mach 3, 6, and 7 on a modified Orion capsule with parametric sinusoidal heat-shield geometry (amplitude A , wave count N), extending Juhany et al. (2024); constituting the first parametric study of sinusoidal modulation on re-entry vehicle heat shield faces.
- Executing a 3×4 full-factorial case matrix to characterise the effects of sinusoidal forward-facing concavities on drag coefficient (C_d) and mean shock standoff distance; post-processing via stagnation streamline line probes in ANSYS Fluent.
- Validated corrected $C_d \approx 1.44$ at Mach 6 vs. baseline (~ 1.30) after diagnosing and resolving a reference-area error.
- Collaborating with Ananya Jahan Chowdhury, Nowroze Farhan Ifraj (Virginia Tech), and Mahir Labib Chowdhury; targeting *Aerospace Science and Technology* or *Results in Engineering* (Q1).

ML-Assisted FSI Analysis of Shockwave–Boundary Layer Interaction in Transonic Flow

2025 – Present

AUST Applied Modelling and Simulation Lab (AASML), Dhaka, Bangladesh

- Investigating aerodynamic flutter of the NACA 64A010 airfoil at $Ma = 0.796$ using two-way FSI, benchmarked against the Isogai (1981) flutter boundary dataset.
- Implementing a density-based solver with SST $k-\omega$ turbulence modelling (compressibility correction), Sutherland viscosity law, and Pressure Far-Field boundary conditions in ANSYS Fluent; structural coupling via ANSYS Static Structural.
- Developing a C-type structured mesh domain; optimising inflation layers and leading-edge sizing for accurate shockwave–boundary layer resolution.
- Machine-learning surrogate for flutter prediction planned as a subsequent phase; collaborating with Tanzim Al Islam, Sumit Saha, and Shraban Sattwik (College of Aviation Technology).

Dynamic Modelling of a District Cooling System (DCS)

2025 – Present

Dept. of Mechanical and Production Engineering, AUST, Dhaka, Bangladesh

- **Supervisor:** Prof. Dr. Dewan Hasan Ahmed
- Developing a component-level dynamic simulation in OpenModelica, modelling chillers, distribution networks, and end-user thermal loads for performance and energy optimisation analysis.

Publications

Conference Proceedings

- **Prithul Das**, Mohammad Nafis Irfan, Salim Subah, Md. Samiul Haider Chowdhury. “Aerodynamic Comparison Analysis of Conic Rocket with Sounder Fin, Shark-Caved Fin, Straight Fin, Trapezoidal Fin in Subsonic Flow.” *Presented at 3rd International Conference on Mechanical Engineering and Applied Sciences (ICMEAS 2025)*, MIST, Dhaka, Bangladesh. July 2025.
- **Prithul Das**, ASM Toufiq Imam, A.F.M. Fahmidur Rahman, Salim Subah, Nazmul Hossain Siam, M. Faisal Riyad.

“Experimental Performance Evaluation of an Earth Tunnel Heat Exchanger (ETHX) System for Space Cooling and Heating in Bangladesh.” *Presented at 6th Annual Paper Meet on Electrical Engineering (APMEE 2025)*, Institution of Engineers, Bangladesh (IEB), Dhaka, Bangladesh. December 2025.

Manuscripts In Progress

- **Prithul Das**, et al. “Aerothermodynamic Analysis of a Modified Orion Capsule Heat Shield with Parametric Sinusoidal Geometry in Hypersonic Flow.” *[In Progress]* — target: *Aerospace Science and Technology* or *Results in Engineering*.
- **Prithul Das**, et al. “Machine-Learning-Assisted FSI Analysis of Shockwave–Boundary Layer Interaction in Transonic Flow over NACA 64A010.” *[In Progress]* — target: *Journal of Fluids and Structures* or *Computers & Fluids*.

Applicable Skills

CAE Tools	ANSYS Fluent, ANSYS Workbench, OpenFOAM (in progress), OpenModelica
CAD Tools	SolidWorks
Programming	Python (NumPy, SciPy, Matplotlib), MATLAB, C
Documentation	LaTeX (technical reports, conference papers, presentations), Microsoft Office

Leadership and Service

Co-Founder

September 2025 – Present

AUST Applied Modelling and Simulation Lab (AASML), Dhaka, Bangladesh

- Co-founded the first student-driven computational research hub at AUST focused on modelling and simulation; operates under two faculty advisors with open membership across the department (aasmlab.org).
- Organised a seminar titled “*Multiscale Flow Physics: Bridging Nano and Micro Domains*” with Maruf Md Ikram (Michigan State University, USA); managing academic sessions and coordinating external speakers.
- Mentoring junior students in ANSYS Fluent workflows, mesh generation, and technical writing; initiating CFD-based research in aerodynamics and thermal-fluid systems.

Communication Director

December 2025 – June 2026

IMechE AUST Student Chapter, Dhaka, Bangladesh

- Lead official communications, technical event publicity, and professional correspondence; coordinate outreach and speaker engagement for chapter events.

Recruitment Director

July – December 2025

IMechE AUST Student Chapter, Dhaka, Bangladesh

- Led recruitment planning, candidate evaluation, and organisational onboarding for new chapter members.

Junior Website and Publicity Director

December 2024 – July 2025

IMechE AUST Student Chapter, Dhaka, Bangladesh

- Managed event publicity and online presence for technical and professional programmes.

Certifications and Courses

- **A Hands-on Introduction to Engineering Simulations** (2024), CornellX / edX — finite-element and CFD simulation fundamentals.
- **OpenFOAM for Beginners** (Ongoing), Flowthermolab — open-source CFD solver workflow and case setup.

Selected Project

- **Vine Robot — Soft Robotics (2025)**: Designed and fabricated a vine robot demonstrating pneumatic actuation using low-cost plastic tubing as a third-year course project.

References

Available upon request.